



Contents lists available at ScienceDirect

Journal of the American Pharmacists Association

journal homepage: www.japha.org

RESEARCH

Long-term opioid therapy in older adults: Incidence and risk factors related to patient characteristics and initial opioid dispensed

Iftekhar Ahmed, Alan J. Zillich, Noll L. Campbell, Kevin M. Sowinski, David R. Foster*



ARTICLE INFO

Article history:

Received 12 August 2024

Accepted 2 December 2024

Available online 10 December 2024

ABSTRACT

Background: The clinical benefits of long-term opioid therapy (LTOT) are not clearly known; however, LTOT is associated with various adverse outcomes.

Objective: This study aimed to estimate the incidence and risk factors associated with LTOT in adults aged ≥ 65 years.

Methods: This was a retrospective cohort study using Medicare claims data. Opioid-naïve older adults filling an opioid prescription between 2014 and 2016 were included. The outcome variable was LTOT, defined as an opioid use episode longer than 90 days and having > 60 cumulative days of supply. Predictor variables included patient characteristics, characteristics of initial opioid dispensed, and pain conditions. Multivariable logistic regression was performed to assess the association between predictors and LTOT.

Results: Among 162,287 participants, 10,296 (6.3%) met the definition of LTOT. Key patient characteristics associated with LTOT were age > 85 years (adjusted odds ratio 1.13 [95% confidence interval 1.05–1.21]); > 5 comorbidities (1.55 [1.45–1.65]); and history of drug use disorder (1.53 [1.35–1.74]), alcohol use disorder (1.38 [1.23–1.54]), tobacco use disorder (1.31 [1.23–1.40]), and opioid use disorder (2.00 [1.69–2.37]). Characteristics of initial opioid associated with LTOT were dispensing long-acting opioids (1.72 [1.21–2.44]) and concomitant use of benzodiazepines (1.16 [1.08–1.25]), gabapentinoids (1.57 [1.47–1.67]), and prescription nonsteroidal anti-inflammatory drugs (1.24 [1.17–1.31]). Anxiety disorders were associated with 1.4–1.5 times increased odds of LTOT. Moreover, initial opioid supply of ≥ 30 days led to 11–16 times higher odds of LTOT than days' supply of 1–3 days.

Conclusions: Factors related to patient characteristics (age, number of comorbidities, substance use disorders, anxiety disorders) and initial opioid dispensation (duration of action, certain concomitant medications, days' supply) are associated with LTOT in older adults. Prescribers should consider these factors when prescribing opioids to senior patients.

© 2024 American Pharmacists Association®. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Background

In the United States, widespread use and misuse of prescription opioids have contributed to the opioid epidemic.¹ From 1999 to 2021, more than 645,000 people died from opioid overdose.² In 2022 alone, more than 14,000 deaths occurred owing to overdose involving prescription opioids.³ Although opioid prescribing rates have decreased in recent years, the average number of days' supply per opioid

prescription and deaths from opioid overdose have increased.⁴ The Centers for Disease Control and Prevention recommends that the initial duration of opioid therapy should be a few days or less for most conditions.⁵ However, many patients receive opioid therapy for a longer period of time (e.g., several months).^{6,7} Although there is a lack of consensus regarding the definition of long-term opioid therapy (LTOT), it is most commonly defined as total opioid days' supply of > 90 days.⁸ Although short to medium duration of opioid therapy is considered to be effective in pain management, the clinical benefits of LTOT in long-term management of noncancer pain are uncertain. No blinded randomized controlled trial has evaluated long-term efficacy of opioids. A meta-analysis of open-label extension trials showed evidence of long-term efficacy of opioids, but the evidence was rated very poor quality.⁹

* **Correspondence:** David R. Foster, PharmD, FCCP, Associate Professor, Department of Pharmacy Practice, College of Pharmacy, Purdue University, FTB Faculty Office Bldg., 640 Eskenazi Ave., Indianapolis, IN 46202.

E-mail address: drfoster@purdue.edu (D.R. Foster).

Key Points

Background:

- Older adults have a higher prevalence of pain than other age groups and are more likely to become long-term opioid users.
- The clinical benefits of long-term opioid therapy (LTOT) are not clearly known; however, LTOT is associated with various adverse outcomes.
- Opioid-related adverse outcomes are often more common in older patients than younger patients, and such adverse outcomes are more likely with LTOT.

Findings:

- Patient-related factors associated with LTOT are greater age, comorbidities, anxiety disorders, history of substance use disorders (drug, tobacco, and alcohol), and history of opioid use disorder.
- Characteristics of initial opioids dispensation predicting LTOT are greater number of days' supply, dispensing long-acting opioids, and use of concomitant benzodiazepines, gabapentinoids, and prescription nonsteroidal anti-inflammatory drugs.
- In patients with certain pain conditions, initiating therapy with tramadol may lead to an increased likelihood of LTOT.

In contrast, LTOT is associated with an increased risk of various adverse clinical outcomes including all-cause mortality, overdose, opioid use disorder (OUD), constipation, fractures, and myocardial infarction.^{5,10-12} In addition, randomized controlled trials have reported mixed evidence regarding the effectiveness of LTOT in quality of life and functional status, with some studies showing improvement, some studies showing no changes, and some studies showing worsening outcomes.¹³ LTOT is also associated with higher health care cost.¹⁴

Compared with the general population, older adults are more likely to receive LTOT. Studies have reported a positive association between age and LTOT.¹⁵⁻¹⁷ Furthermore, opioid-related adverse outcomes are often more common in older patients than younger patients,¹⁸⁻²⁰ and such adverse outcomes may be exacerbated by long-term use of opioids. Therefore, experts generally recommend avoiding LTOT in older adults unless absolutely necessary. To successfully decrease the incidence of LTOT, it is important to understand which factors increase the risk of LTOT in seniors. However, risk factors for LTOT in older adults are not well characterized.

Objectives

This study has 2 objectives: (1) to determine the incidence of LTOT in older adults and (2) to identify the risk factors of LTOT related to patient characteristics, characteristics of initial opioid dispensed, and pain conditions for which the initial opioid was prescribed.

Methods

Data source

Medicare claims data from 2014 to 2016 were used in this study. Limited data were obtained from Research Data Assistance Center, which is a Centers for Medicare and Medicaid Services (CMS) contractor and provides services related to the access and use of CMS data.²¹ A nonrandom sample of 1 million patients obtained for use in a separate study (unrelated to opioid use) was used for this study.²² All patients in the dataset were alive during the entire duration of study. Demographic information (age, sex, race) was obtained from the Master Beneficiary Summary File (MBSF) Base Claims dataset. The total number of comorbidities and specific comorbidities such as depression, anxiety disorders, and substance use disorders were identified from the MBSF Chronic Conditions Segment and the MBSF Other Chronic or Potentially Disabling Conditions Segment datasets. These chronic condition datasets are based on the chronic conditions data warehouse developed by CMS which uses specific algorithms based on ICD codes to identify these conditions.²³ The Part D Event File was used to extract information on opioid dispensing and characterize initial opioid use and determine LTOT. To identify the pain conditions at the time of initial opioid use, the Base Claim Files (inpatient, outpatient, carrier, skilled nursing facility, and home health agency) were used. The study was approved by the Purdue University Institutional Review Board.

Sample

The study population was opioid-naïve older adults (65–105 years) receiving an opioid prescription for acute or chronic noncancer pain. Patients were included if they fulfilled the following inclusion criteria: (1) had an opioid prescription filled between 2014 and 2016, (2) were opioid naïve (did not fill an opioid prescription for at least 6 months before the initial opioid use date), and (3) had at least 6 months of available data after the last opioid fill. Participants were excluded if they were diagnosed as having cancer (except nonmelanoma skin cancer), were receiving palliative care, or were receiving medications for the treatment of OUD (buprenorphine [Subutex, Probuphine, Sublocade], buprenorphine-naloxone, pentazocine HCl-naloxone HCl).

Outcome variable

The outcome variable was LTOT. We defined LTOT using an episodic approach as illustrated in Figure 1. The start of an episode was defined as the day any opioid prescription was filled after a period of at least 6 months without an opioid prescription being filled. The initial opioid was the first opioids filled in an episode. The last opioid fill in an episode was marked by an absence of any opioid fills in the subsequent 6 months. The duration of an episode was defined as the period between the first and the last opioid fill plus the number of days' supply for the last fill. The cumulative days of supply were calculated by adding the total days' supply of all opioids filled during an episode. An episode was considered long-term if it lasted longer than 90 days and had more than 60 cumulative days' supply. This is a modified version of the definition

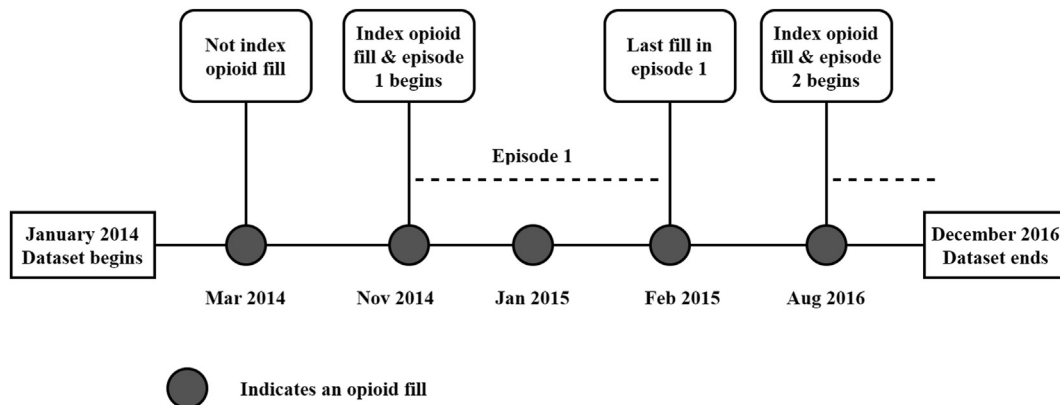


Figure 1. Timeline (hypothetical) showing how an episode of opioid use was defined in this study. In this hypothetical example, the index or initial opioid fill and start of an episode is defined as the day any opioid prescription is filled after a period of at least 6 months without an opioid prescription being filled. Given that only 2 months of data is available before the fill in March 2014, this is not considered an initial fill. The next prescription is filled in November 2014. It is considered an initial fill given that there is an 8-month gap between the last fill (March 2014) and this one. The last opioid fill in an episode is marked by an absence of any opioid fill in the subsequent 6 months. The fill in February 2015 fulfills this condition and marks the end of episode 1. The fill in August 2016 marks the beginning of the second episode (not included in our study).

developed by the Consortium to Study Opioid Risks and Trends (CONSORT).^{15,24,25} In our study, the criteria for the duration of the episode were the same as that of CONSORT. However, we changed the criteria for the cumulative days' supply from 120 days to 60 days in consideration of recent guidelines recommending shorter duration of opioid therapy and recent studies using more stringent criteria to define LTOT.^{5,8,26} If a patient had multiple episodes of opioid use during the study period,

(gabapentin, pregabalin), selective serotonin reuptake inhibitors (SSRIs) or serotonin and norepinephrine reuptake inhibitors (SNRIs), and prescription nonsteroidal anti-inflammatory drugs (NSAIDs). Medications were considered concomitant if they were dispensed within 30 days before or after the dispensing date of the initial opioid. The morphine milligram equivalents (MMEs) per day for all opioids were calculated using this formula:

$$\text{Strength per unit} \times (\text{number of units} / \text{days' supply}) \times \text{MME conversion factor} = \text{MME} / \text{day}$$

only the first episode was considered. The list of all opioid-related medications is in [Appendix 1](#).

Predictor variables

We included 3 groups of predictor variables: patient characteristics, characteristics of initial opioid dispensed, and pain conditions at the time of dispensing. Patient characteristics included age, sex, race, number of comorbidities, depression (including bipolar and other depressive mood disorders) within the last 5 years, anxiety disorders within the last 5 years, and history of substance use disorders such as drug use disorder (DUD), alcohol use disorder (AUD), tobacco use disorder (TUD), and OUD.

The variables related to initial opioid dispensation were the type of opioid dispensed (hydrocodone, oxycodone, codeine, tramadol, or others), duration of action of opioids (short acting vs long acting), number of days' supply, average daily dose, and concomitant medications. Opioid medications were identified using the National Drug Codes. We also identified concomitant use of prescription benzodiazepines ("Z-drugs" [eszopiclone, zaleplon, zolpidem] were not included), gabapentinoids

The MME conversion factors were based on the guidelines published by the U.S. Department of Health and Human Services.²⁷ When multiple opioids were dispensed during the initial dispensation, the longest days' supply was considered and the average daily doses in MME for all the opioids were summated.

Pain conditions were identified using ICD-9 and ICD-10 diagnosis codes and organized into 7 categories: back and neck pain, osteoarthritis and joint pain, limb or extremity pain, trauma and surgery, abdominal and bowel pain, neuropathic pain and fibromyalgia, and other pain conditions.^{7,28} Pain conditions grouped as other pain conditions included dental pain, nonspecific chronic pain, bone infections, rheumatoid arthritis, infectious arthritic diseases, spinal cord injury, complex regional pain syndrome, other inflammatory arthropathies, sickle cell disease, systemic lupus erythematosus, acquired deformities (excluding back conditions), and urogenital, pelvic, and menstrual pain. If an encounter with the aforementioned conditions happened within 30 days before an opioid was dispensed, it was considered an existing pain condition for that patient. Patients could have multiple pain diagnoses.

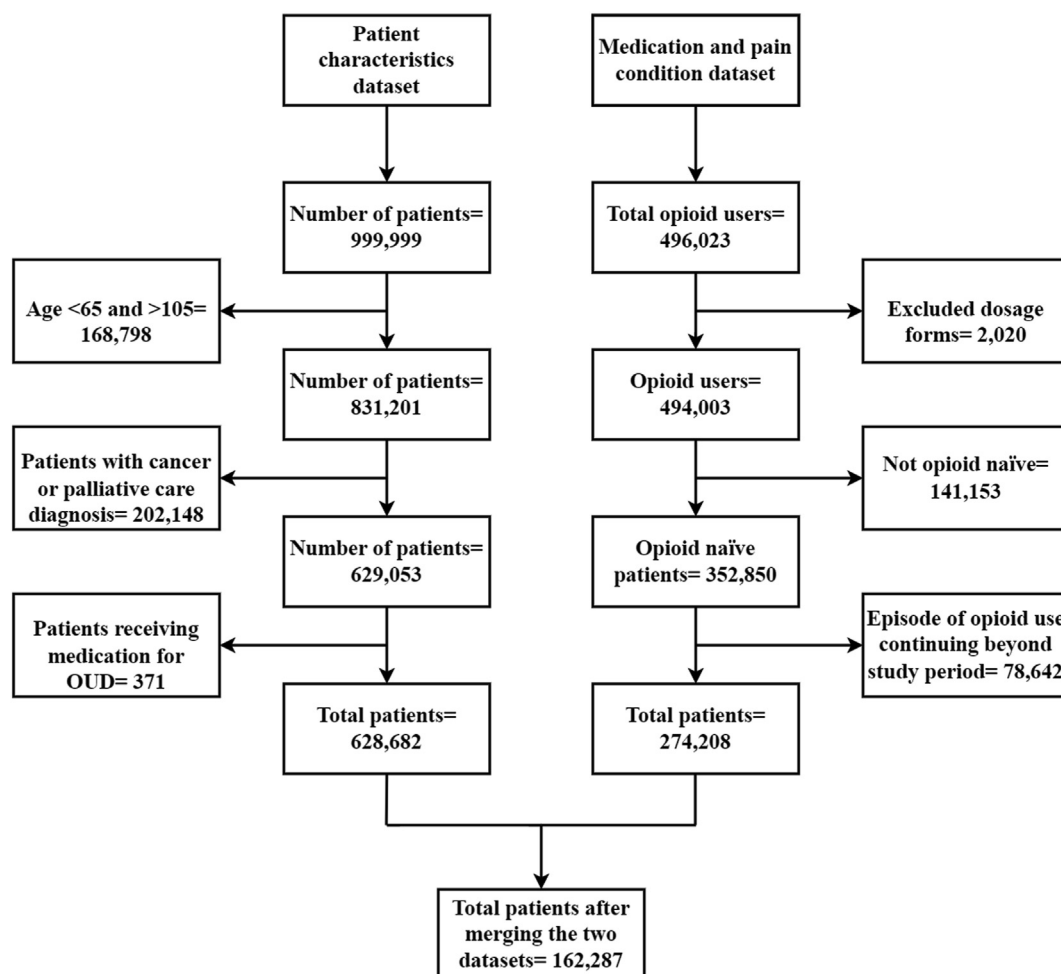


Figure 2. Flow diagram of patient selection. The patient characteristics dataset contained information on demographics, comorbidities, and substance use disorders. The medication and pain condition dataset contained information on dispensing of opioids and other medications and pain conditions. Both datasets were created by merging several primary Medicare datasets (details in the Methods section). Abbreviation used: OUD, opioid use disorder.

Statistical analysis

We used descriptive analysis to report the frequencies of patient characteristics, characteristics of initial opioid dispensed, and pain conditions. To compare the incidence of LTOT across various groups and to compare the characteristics of long-term opioid users and short-term opioid users, chi-square tests were performed.

We conducted multivariable logistic regression to identify the factors associated with LTOT. The analyses were performed in 2 steps. First, we ran multivariable logistic regression with backward elimination (significance level = 0.05) using all of the main effect variables and clinically relevant interaction effect variables. This was a preliminary step to identify statistically significant interactions. In the second step, we performed a full model multivariable logistic regression with all main effect variables and the statistically significant interaction effect variables from the previous step. The adjusted odds ratios (AOR) along with 95% confidence intervals (CI) of main effect variables were reported only if they were not involved in any interaction. In both steps, the regression model was adjusted for all variables.

The effect sizes for variables such as anxiety disorders, depression, pain conditions, type of opioid dispensed, number of days' supply, average daily dose, and concomitant SSRIs or SNRIs could not be estimated in the primary analysis given that they were involved in statistical interactions. Therefore, to estimate their effect sizes, we performed secondary analyses after stratifying the dataset. Given that prescription for opioids may vary depending on the pain conditions and because pain conditions had statistically significant interactions with several variables, we stratified the dataset based on the 3 most common pain condition categories (back and neck pain, osteoarthritis and joint pain, and abdominal and bowel pain). Then, we estimated the odds of LTOT by the presence of anxiety disorders, depression, type of opioid dispensed, number of days' supply, average daily dose, and concomitant SSRI or SNRI for each subgroup using multivariable logistic regression adjusting for all variables. In the secondary analysis, we used methods similar to the 2-step analysis mentioned in the primary analysis. For all analyses, a *P* value less than 0.05 was considered statistically significant. Statistical analyses were performed in SAS 9.4 (SAS Institute, Cary, NC). We also

performed a sensitivity analysis by changing one of the 2 criteria for defining LTOT (90 cumulative days' supply instead of 60 cumulative days' supply).

Results

A total of 162,287 patients met the inclusion criteria (Figure 2). Descriptive data on patient characteristics, initial opioid characteristics, and pain conditions are presented in Table 1. The mean age of the patients was 74.7 years (SD 7.2), and most patients were female (63.2%). Short-acting hydrocodone was the most prescribed opioid, followed by short-acting tramadol, short-acting oxycodone, and codeine. Approximately 13% received > 50 MME of opioids, and 10% had days' supply of 30 days or longer. The most common pain condition was osteoarthritis and joint pain, followed by back and neck pain and abdominal and bowel pain. A comparison of the characteristics between long-term opioid users and short-term opioid users is in Appendix 1.

Incidence of LTOT

The incidence of LTOT was 6.3% ($n = 10,296$). The comparative incidences of LTOT in various groups are presented in Table 1. Females, participants > 85 years, and participants who were black had a higher incidence of LTOT. Moreover, patients having > 5 comorbidities, depression, anxiety disorders, and a history of substance use disorders had higher rates of LTOT. Regarding the characteristics of initial opioid dispensation, patients receiving tramadol, long-acting opioids, ≤ 20 MME, ≥ 30 days' supply of opioids, and concomitant psychotropics and analgesics had the highest rates of LTOT. Among the various pain conditions, the highest incidence of LTOT was observed among subjects with back and neck pain, followed by osteoarthritis and joint pain, and neuropathic pain and fibromyalgia.

Relationship between patient characteristics and LTOT

Patient-related factors associated with LTOT are presented in Table 2. Demographic characteristics associated with LTOT were age older than 85 years (AOR 1.13 [95% CI 1.05–1.21]) and being black (1.11 [1.01–1.21]). Patients with more than 5 comorbidities had 1.55 times higher odds (95% CI 1.45–1.65) of receiving LTOT than patients with 0–2 comorbidities. Risk factors related to substance use disorders included history of DUD (1.53 [1.35–1.74]), history of AUD (1.38 [1.23–1.54]), and history of TUD (1.31 [1.23–1.40]). Having a history of OUD doubled the odds of LTOT (2.00 [1.69–2.37]).

The effect sizes of depression and anxiety disorders could not be estimated for the whole cohort given that they were involved in statistical interactions. Therefore, in the secondary analyses, the cohort was stratified into 3 subgroups based on the most common pain conditions, and the effect sizes of depression and anxiety disorders were estimated for each subgroup (Table 3). In 2 of the 3 subgroups (back and neck pain, osteoarthritis and joint pain), depression was not associated with LTOT. Alternatively, in all 3 subgroups, anxiety disorders led to 1.36–1.50 times higher odds of LTOT.

Relationship between characteristics of initial opioid dispensed and LTOT

Patients who were dispensed long-acting opioids were 1.72 times more likely to receive LTOT (AOR 1.72 [95% CI 1.21–2.44]) (Table 2). Concomitant uses of benzodiazepines (1.16 [1.08–1.25]) and gabapentinoids (1.57 [1.47–1.67]) were significantly associated with LTOT. Similarly, concomitant prescription of NSAIDs increased the odds of LTOT by 24% (1.24 [1.17–1.31]).

Similar to the patient characteristics, the effect sizes of some of the opioid dispensation-related variables including pain condition, opioid type, number of days' supply, average daily dose, and concomitant SSRI or SNRI could not be estimated for the whole cohort given that they were involved in statistical interactions. Therefore, in the secondary analyses, the effect sizes of these variables were estimated for each of the 3 pain subgroups (Table 3). Initiating therapy with tramadol increased the odds of LTOT compared with hydrocodone but the results were statistically significant only in patients with osteoarthritis and joint pain (AOR 1.23 [95% CI 1.07–1.41]) and abdominal and bowel pain (1.50 [1.03–2.19]). The number of days' supply was positively associated with LTOT for all pain subgroups. Compared with initial days' supply of 1–3 days, opioid supply for 30 days or more increased the odds of LTOT by approximately 11- to 16-fold. In all pain types, there was no difference in odds of LTOT between the lowest and highest dose categories. The use of concomitant SSRIs or SNRIs led to an increased odds of LTOT in all 3 pain subgroups, but the increase was statistically significant only in osteoarthritis and joint pain (AOR 1.28 [96% CI 1.08–1.50]).

Sensitivity analyses using a different criterion to define LTOT (90 days' supply instead of 60 days') yielded results that were mostly consistent with the main analysis (Appendix 1).

Discussion

In the present study, we have identified the incidence and risk factors of LTOT in older adults. To date, very few studies have assessed the risk factors for LTOT in older adults even though older adults are at high risk of LTOT and associated adverse outcomes.^{15–20} The incidence of LTOT in our study population was 6.34%. The incidences of LTOT in other studies evaluating older adults range from 5% to 15.5%.^{29–33} In contrast, studies conducted on the general population have generally reported a lower incidence of LTOT (1.3%–6.94%) than that of older adults.^{6,34–37} This higher incidence of LTOT in older adults than the general population may be attributed to a higher prevalence of chronic and persistent pain among older adults. The wide range in the incidence of LTOT in different studies can be explained by the fact that the studies often vary in terms of age and pain conditions of the study populations, but, more importantly, the criteria used to define LTOT. To ensure a robust definition for LTOT, we used a combination of 2 different criteria to define LTOT (episode > 90 days and cumulative days' supply > 60 days). Other studies have used various criteria to define LTOT such as the number of days' supply (with cutoff > 30, > 60, or > 90 days), the number of prescriptions, treatment episode, or a combination of 2 different criteria.⁸ Regardless of the definition used, our study

Table 1
Baseline characteristics of the study cohort and the incidence of LTOT

Variables	No. participants, n (%) (total = 162,287)	Incidence of LTOT, % (overall incidence = 6.3%)	
		Incidence, %	P value ^a
Patient characteristics			
Age (y), mean (SD)	74.68 (7.16)		
Age categories			< 0.001
65–75	98,983 (60.99)	5.65	
76–85	47,646 (29.36)	6.97	
> 85	15,658 (9.65)	8.84	
Sex			< 0.001
Male	59,739 (36.81)	5.27	
Female	102,548 (63.19)	6.97	
Race			< 0.001
White	147,264 (90.74)	6.25	
Black	7375 (4.54)	7.96	
Other	6064 (3.74)	7.04	
Unknown	1584 (0.98)	4.67	
Comorbidities ^b			< 0.001
0–2	43,802 (26.99)	3.92	
3–5	73,640 (45.38)	5.92	
> 5	44,845 (27.63)	9.41	
Depression			< 0.001
No	136,810 (84.30)	5.92	
Yes	25,477 (15.70)	8.64	
Anxiety disorders			< 0.001
No	140,040 (86.29)	5.84	
Yes	22,247 (13.71)	9.50	
History of DUD			< 0.001
No	159,510 (98.29)	6.18	
Yes	2777 (1.71)	15.63	
History of AUD			< 0.001
No	157,985 (97.35)	6.24	
Yes	4302 (2.65)	10.25	
History of TUD			< 0.001
No	147,117 (90.65)	6.07	
Yes	15,170 (9.35)	8.98	
History of OUD			< 0.001
No	161,024 (99.22)	6.24	
Yes	1263 (0.78)	19.56	
Characteristics of initial opioid dispensed and pain conditions			
Opioid type			< 0.001
Hydrocodone SA	73,731 (45.43)	4.75	
Tramadol SA	41,110 (25.33)	11.07	
Oxycodone SA	28,524 (17.58)	4.25	
Codeine	14,856 (9.15)	3.93	
Multiple opioids	1932 (1.19)	8.59	
Others	2134 (1.31)	13.40	
Duration of action of opioid			< 0.001
Short acting	161,065 (99.25)	6.24	
Long acting	745 (0.46)	25.64	
Short acting and long acting	477 (0.29)	13.00	
Average daily dose in MME			< 0.001
≤ 20	56,784 (34.99)	10.65	
21–50	84,777 (52.24)	4.05	
> 50	20,726 (12.77)	3.94	
No. days' supply			< 0.001
1–3	45,840 (28.25)	1.78	
4–7	58,934 (36.31)	3.08	
8–14	25,663 (15.81)	6.34	
15–29	15,093 (9.30)	12.91	
≥ 30	16,757 (10.33)	24.41	

Table 1 (continued)

Variables	No. participants, n (%) (total = 162,287)	Incidence of LTOT, % (overall incidence = 6.3%)	
		Incidence, %	P value ^a
Concomitant benzodiazepine and gabapentinoids			< 0.001
None	134,530 (82.90)	5.55	
Benzodiazepine only	14,089 (8.68)	8.06	
Gabapentinoids only	12,025 (7.41)	12.18	
Benzodiazepine and gabapentinoids	1643 (1.01)	14.36	
Concomitant SSRIs or SNRIs			< 0.001
No	137,730 (84.87)	5.83	
Yes	24,557 (15.13)	9.21	
Concomitant prescription NSAIDs			< 0.001
No	135,937 (83.76)	6.08	
Yes	26,350 (16.24)	7.73	
Pain conditions			< 0.001
Back and neck pain	13,279 (8.18)	9.90	
Osteoarthritis and joint pain	15,327 (9.44)	8.49	
Limb or extremity pain	3589 (2.21)	4.68	
Abdominal and bowel pain	8358 (5.15)	2.55	
Neuropathic pain and fibromyalgia	1924 (1.19)	8.37	
Trauma and surgery	5605 (3.45)	3.32	
Other pain conditions ^c	2953 (1.82)	6.43	
Multiple pain conditions	62,209 (38.33)	7.21	
Unknown	49,043 (30.22)	4.64	

Abbreviations used: AUD, alcohol use disorder; CMS, Centers for Medicare and Medicaid Services; DUD, drug use disorder; MME, morphine milligram equivalent; NSAID, nonsteroidal anti-inflammatory drug; OUD, opioid use disorder; SA, short acting; SSRI, selective serotonin reuptake inhibitor; SNRI, serotonin and norepinephrine reuptake inhibitor; TUD, tobacco use disorder.

^a P value of chi-square tests of the differences in incidence of LTOT across various groups.

^b Number of comorbidities was calculated from the Master Beneficiary Summary File Chronic Conditions Segment dataset, which is based on the chronic conditions data warehouse developed by CMS.

^c Other pain conditions include dental pain, nonspecific chronic pain, bone infections, rheumatoid arthritis, infectious arthritic diseases, spinal cord injury, complex regional pain syndrome, other inflammatory arthropathies, sickle cell disease, systemic lupus erythematosus, acquired deformities (excluding back conditions), and urogenital, pelvic, and menstrual pain.

supported other findings that increasing age is a risk factor for LTOT.

We analyzed whether the characteristics of the initial opioid were associated with the overall duration of opioid use in opioid-naïve patients. Among the various characteristics of initial opioid dispensation, the number of days' supply was the strongest predictor of LTOT in this study and we observed a dose-response relationship between days' supply and LTOT. In our review of the literature, we identified only one study assessing the impact of initial days' supply of opioids on LTOT in older adults and that study also found an increased risk of LTOT with a higher initial days' supply of opioids.³⁸ The effect of the average daily dose of initial opioids on LTOT was inconsistent in this study, and initial dose of > 50 MME did not result in increased odds of LTOT compared with lower daily doses. The only other study investigating the effect of initial

Table 2

Results of multivariable logistic regression showing the association of patient characteristics and characteristics of initial opioid dispensed with LTOT

Variables	Adjusted OR (95% CI)
Patient characteristics	
Age	
65–75	1
76–85	1.03 (0.98–1.08)
> 85	1.13 (1.05–1.21)
Sex	
Male	1
Female	1.05 (1.00–1.10)
Race	
White	1
Black	1.11 (1.01–1.21)
Other	0.89 (0.80–0.99)
Comorbidities	
0–2	1
3–5	1.21 (1.14–1.29)
> 5	1.55 (1.45–1.65)
History of DUD	
No	1
Yes	1.53 (1.35–1.74)
History of AUD	
No	1
Yes	1.38 (1.23–1.54)
History of TUD	
No	1
Yes	1.31 (1.23–1.40)
History of OUD	
No	1
Yes	2.00 (1.69–2.37)
Characteristics of initial opioid dispensed	
Duration of action of opioid	
Short acting	1
Long acting	1.72 (1.21–2.44)
Short acting and long acting	1.31 (0.88–1.95)
Concomitant benzodiazepine and gabapentinoids	
None	1
Benzodiazepine only	1.16 (1.08–1.25)
Gabapentinoids only	1.57 (1.47–1.67)
Benzodiazepine and gabapentinoids	1.69 (1.45–1.96)
Concomitant prescription NSAIDs	
No	1
Yes	1.24 (1.17–1.31)

Abbreviations used: AUD, alcohol use disorder; CI, confidence interval; DUD, drug use disorder; NSAID, nonsteroidal anti-inflammatory drug; OR, odds ratio; OUD, opioid use disorder; TUD, tobacco use disorder.

Note: This table presents results for the whole cohort and includes variables that were not involved in statistical interactions. The statistically significant interactions were pain condition × number of days' supply, pain condition × dose, opioid type × number of days' supply, opioid type × dose, number of days' supply × dose, and depression × concomitant SSRIs or SNRIs. ORs for variables such as opioid type, number of days' supply, average daily dose, concomitant SSRIs or SNRIs, anxiety, and depression have been reported for specific pain conditions (Table 3) given that they were involved in interactions.

daily dose reported that initial daily dose of > 100 MME increased the risk of LTOT in older adults.²⁹ It is well established that a higher dose of opioids is associated with an increased risk of overdose and opioid dosage of > 50 MME is not generally recommended.⁵ The association of concomitant use of gabapentinoids, SSRIs or SNRIs, and prescription NSAIDs with LTOT in older adults had not been previously studied. We found that the use of concomitant medications such as benzodiazepines, gabapentinoids, and prescription NSAIDs was

very common, and all of those medications increased the likelihood of LTOT. There are also risks of various adverse outcomes with concomitant use of opioids, benzodiazepines, and gabapentinoids.^{5,39,40} Whether concomitant use of SSRIs or SNRIs increases the likelihood of LTOT in older adults is unknown. Ours is among the first studies to assess the risk of LTOT in older adults using concomitant SSRIs or SNRIs, and we observed inconsistent results across the 3 pain subgroups. More research is needed to clearly understand the role of concomitant use of SSRIs or SNRIs in LTOT.

We observed that for osteoarthritis and joint pain as well as abdominal and bowel pain, the selection of tramadol as the initial opioid resulted in higher odds of LTOT than hydrocodone. Our finding regarding tramadol is supported by a study of the general population where initiating therapy with tramadol increased the risk of LTOT.⁷ Only one study of senior patients compared various opioids in terms of the risk of LTOT in seniors and found that tramadol, oxycodone, and codeine posed similar risks of LTOT and all of them posed an increased risk of LTOT compared with morphine.⁴¹ We also observed in our study that the crude incidence of LTOT for tramadol was more than double that of hydrocodone or oxycodone. Further analysis revealed that the initial days' supply was much higher for tramadol than hydrocodone or oxycodone. For example, nearly 22% of tramadol users had an initial days' supply of > 30 days compared with only 4%–6% of patients receiving hydrocodone and oxycodone. This suggests that tramadol may be more commonly used for long-term management of pain. This may be explained by a perception among clinicians that tramadol is less addictive and “safer” than other opioids.⁴² Another reason for increased use of tramadol for long-term use may be its classification as a schedule IV opioid whereas hydrocodone and oxycodone are schedule II opioids.⁴³ The rescheduling of hydrocodone in 2014 led to an increase in prescriptions for other opioids, especially tramadol.⁴⁴ Tramadol's apparent popularity is also supported by the fact that during the period of our study (2014–2016), an increased rate of tramadol prescription was observed in the United States.⁴²

Also in our study, anxiety disorders were associated with LTOT, which is consistent with another study on older adults.⁴⁵ In contrast, depression was not a statistically significant risk factor for LTOT in 2 of the 3 pain conditions. We did not find any study of older adults that investigated the association between depression and LTOT. Studies in the general population have reported conflicting evidence, with some studies reporting a positive association between depression and LTOT^{46,47} whereas others showing no association.^{48,49} Future studies should further investigate the role of depression in LTOT. Of note, many of the risk factors of LTOT identified in this study including anxiety, OUD, SUDs, TUD, long-acting opioids, and benzodiazepines are also associated with opioid-induced respiratory depression (OIRD).^{50,51} Although opioid dose and depression were not significantly associated with LTOT in this study, they have been found to be associated with OIRD in other studies.^{50,51}

Strengths and limitations

The primary strengths of our study are the large sample size and the ability to answer our research question using real-world evidence. Moreover, the use of secondary claims data

Table 3

Results of multivariable logistic regression for specific pain condition subgroups showing the association of patient characteristics and characteristics of initial opioid dispensed with LTOT

Variables	Back and neck pain Adjusted OR (95% CI)	Osteoarthritis and joint pain Adjusted OR (95% CI)	Abdominal and bowel pain Adjusted OR (95% CI)
Patient characteristics			
Depression			
No	1	1	1
Yes	1.11 (0.94–1.31)	1.00 (0.84–1.18)	1.68 (1.16–2.42)
Anxiety disorders			
No	1	1	1
Yes	1.36 (1.15–1.60)	1.45 (1.22–1.71)	1.50 (1.01–2.19)
Characteristics of initial opioid dispensed			
Opioid type			
Hydrocodone SA	1	1	1
Codeine	1.0 (0.76–1.30)	0.96 (0.74–1.24)	0.55 (0.22–1.16)
Oxycodone SA	0.92 (0.73–1.15)	0.69 (0.53–0.90)	0.85 (0.54–1.33)
Tramadol SA	1.13 (0.98–1.30)	1.23 (1.07–1.41)	1.50 (1.03–2.19)
Other opioids	1.09 (0.46–2.27)	1.33 (0.60–2.61)	2.60 (1.04–5.74)
Multiple opioids	1.69 (0.82–3.24)	0.38 (0.14–0.83)	2.29 (0.30–10.5)
No. days' supply			
1–3	1	1	1
4–7	1.70 (1.27–2.31)	2.03 (1.49–2.82)	0.85 (0.56–1.30)
8–14	2.97 (2.21–4.04)	4.01 (2.93–5.58)	1.60 (0.91–2.76)
15–29	5.83 (4.33–7.95)	7.99 (5.82–11.17)	4.81 (2.63–8.64)
≥ 30	11.52 (8.61–15.66)	15.97 (11.68–22.28)	14.49 (8.65–24.51)
Average daily dose (MME)			
≤ 20	1	1	1
21–50	1.26 (1.08–1.48)	1.26 (1.07–1.48)	1.07 (0.72–1.61)
> 50	1.11 (0.79–1.53)	1.04 (0.74–1.45)	0.65 (0.31–1.31)
Concomitant SSRIs or SNRIs			
No	1	1	1
Yes	1.13 (0.96–1.33)	1.28 (1.08–1.50)	1.04 (0.70–1.51)

Abbreviations used: CI, confidence interval; MME, morphine milligram equivalent; OR, odds ratio; SA, short acting; SNRI, serotonin and norepinephrine reuptake inhibitor; SSRI, selective serotonin reuptake inhibitor.

Note: This table only includes variables that had interactions in the analysis of the whole dataset.

allowed us to study long-term opioid use, which is not commonly studied in clinical trials. Second, Medicare data have several strengths. It represents more than 55 million U.S. individuals as of 2015 and contains comprehensive records of health care services in all care settings, and very rarely do patients withdraw from Medicare, which means there is very little loss to follow-up.⁵² For defining LTOT, we used a combination of 2 definitions (treatment episode and number of days' supply), which is a more robust approach to capturing LTOT than some prior studies.⁸ Our study also has limitations. The study used a nonrandom sample of the Medicare population, which was originally intended for another study.²² However, the general characteristics (age, sex, race) of this sample were similar to the Medicare population (Appendix 1). Although rigorous statistical methods were used to control for the effect of confounders, this does not rule out the possibility of confounding by unmeasured or incompletely measured variables. Moreover, we could not determine the pain conditions for approximately 30% of the patients (however, this is fairly common in studies of claims data).^{7,53} We relied on ICD codes for identification of pain conditions and other variables, which may lead to misclassification of diseases. In addition, this dataset does not capture the NSAIDs purchased over the counter, and our assessment of NSAIDs is limited to only prescription NSAIDs. The age of the dataset may be a potential limitation. Data are from 2014 to 2016 and opioid prescribing patterns may have changed since that time. However, we

speculate that any prescribing pattern changes would have a greater impact on the estimate of the frequency of LTOT rather than the risk factors associated with LTOT.

Significance of the findings

Older adults have a higher prevalence of chronic pain than the general population⁵⁴ and are more likely to receive opioid prescriptions, including potentially inappropriate opioid prescriptions.^{19,55} Decrease in hepatic and renal function in older adults reduces drug metabolism and renal clearance, which may lead to higher risks of opioid-related adverse events such as overdose or respiratory depression. Multimorbidity and polypharmacy are very common in the older population, and many opioid users also use concomitant medications, which can affect the duration and outcomes of opioid therapy. Understanding patient level and dispensation-related factors associated with LTOT may allow clinicians to identify patients who are at higher risk of transitioning to LTOT and avoid potentially risky prescribing. Before prescribing opioids, clinicians should consider potential factors that increase the likelihood of a patient progressing to LTOT. Prescribers need to be especially careful when prescribing opioids to patients with multiple comorbidities, anxiety disorders, and history of substance use disorders and OUD given that these patients have higher odds of LTOT according to our findings. Moreover, our study has

shown that patterns of initial opioid therapy are associated with LTOT and careful selection of the opioid regimen is very crucial when prescribing opioid to an opioid-naïve patient. To minimize the risk of LTOT, prescribers should start the therapy with short-acting opioids and shorter days' supply and try to avoid concomitant prescription of medications that increase the risk of LTOT. Understandably, LTOT may be necessary in certain situations, and clinicians should evaluate the risks and benefits of LTOT before initiating LTOT in older adults. Nevertheless, unintentional use of LTOT should be avoided, and being aware of the aforementioned risk factors may help.

Conclusions

We have identified several factors related to patient characteristics and initial opioid dispensation predicting LTOT in older adults. Understanding these risk factors may enable prescribers to adopt safer prescribing practices that may prevent LTOT in older adults. Future studies should further investigate strategies to limit LTOT among older adults.

Disclosure

The authors declare no relevant conflicts of interest or financial relationships.

Acknowledgments

We acknowledge and thank Dr. Chong Gu and David Rowe from the Department of Statistics at Purdue University for their assistance with the statistical analyses.

Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.japh.2024.102311>.

References

- Skolnick P. The opioid epidemic: crisis and solutions. *Annu Rev Pharmacol Toxicol*. 2018;58(1):143–159.
- Centers for Disease Control and Prevention. Understanding the opioid overdose epidemic; 2024. Available at: <https://www.cdc.gov/overdose-prevention/about/understanding-the-opioid-overdose-epidemic.html>. Accessed July 14, 2024.
- National Institute on Drug Abuse. Drug overdose death rates. Available at: <https://nida.nih.gov/research-topics/trends-statistics/overdose-death-rates>. Accessed July 1, 2024.
- Centers for Disease Control and Prevention. Annual surveillance report of drug-related risks and outcomes—United States; 2019. Available at: <https://stacks.cdc.gov/view/cdc/84281>. Accessed July 1, 2024.
- Dowell D, Ragan KR, Jones CM, Baldwin GT, Chou R. CDC clinical practice guideline for prescribing opioids for pain—United States, 2022. *MMWR Recomm Rep*. 2022;71(3):1–95.
- Hayden JA, Ellis J, Asbridge M, et al. Prolonged opioid use among opioid-naïve individuals after prescription for nonspecific low back pain in the emergency department. *Pain*. 2021;162(3):740–748.
- Shah A, Hayes CJ, Martin BC. Factors influencing long-term opioid use among opioid naïve patients: an examination of initial prescription characteristics and pain etiologies. *J Pain*. 2017;18(11):1374–1383.
- Shen Y, Bhagwandass H, Branchcomb T, et al. Chronic opioid therapy: a scoping literature review on evolving clinical and scientific definitions. *J Pain*. 2021;22(3):246–262.
- Bialas P, Maier C, Klose P, Häuser W. Efficacy and harms of long-term opioid therapy in chronic non-cancer pain: systematic review and meta-analysis of open-label extension trials with a study duration ≥ 26 weeks. *Eur J Pain*. 2020;24(2):265–278.
- Chou R, Turner JA, Devine EB, et al. The effectiveness and risks of long-term opioid therapy for chronic pain: a systematic review for a National Institutes of Health Pathways to Prevention Workshop. *Ann Intern Med*. 2015;162(4):276–286.
- Baldini A, Von Korff M, Lin EH. A review of potential adverse effects of long-term opioid therapy: a practitioner's guide. *Prim Care Companion CNS Disord*. 2012;14(3):PCC.11m01326.
- Saunders KW, Dunn KM, Merrill JO, et al. Relationship of opioid use and dosage levels to fractures in older chronic pain patients. *J Gen Intern Med*. 2010;25(4):310–315.
- Sehgal N, Colson J, Smith HS. Chronic pain treatment with opioid analgesics: benefits versus harms of long-term therapy. *Expert Rev Neurother*. 2013;13(11):1201–1220.
- Johnson A, Milne B, Jamali N, et al. Chronic opioid use after joint replacement surgery in seniors is associated with increased healthcare utilization and costs: a historical cohort study. *Can. J. Anaesth*. 2022;69(8):963–973.
- Campbell CI, Weisner C, LeResche L, et al. Age and gender trends in long-term opioid analgesic use for noncancer pain. *Am J Public Health*. 2010;100(12):2541–2547.
- Fritz JM, King JB, McAdams-Marx C. Associations between early care decisions and the risk for long-term opioid use for patients with low back pain with a new physician consultation and initiation of opioid therapy. *Clin J Pain*. 2018;34(6):552–558.
- Ray GT, Bahorik AL, VanVeldhuisen PC, Weisner CM, Rubinstein AL, Campbell CI. Prescription opioid registry protocol in an integrated health system. *Am J Manag Care*. 2017;23(5):e146–e155.
- Papaleontiou M, Henderson CR, Turner BJ, et al. Outcomes associated with opioid use in the treatment of chronic noncancer pain in older adults: a systematic review and meta-analysis. *J Am Geriatr Soc*. 2010;58(7):1353–1369.
- Rose AJ, McBain R, Schuler MS, et al. Effect of age on opioid prescribing, overdose, and mortality in Massachusetts, 2011 to 2015. *J Am Geriatr Soc*. 2019;67(1):128–132.
- West NA, Dart RC. Prescription opioid exposures and adverse outcomes among older adults. *Pharmacoepidemiol Drug Saf*. 2016;25(5):539–544.
- Research Data Assistance Center (ResDAC); 2023. Available at: <https://www.cms.gov/data-research/research/general-information/research-data-assistance-center-resdac>. Accessed July 31, 2024.
- Waghmare PH, Huang C-Y, Jaynes HA, et al. Disparities in Medicare beneficiaries' receiving medication synchronization. *J Am. Pharm. Assoc*. (2003). 2023;63(2):538–546.e2.
- Centers for Medicare & Medicaid Services chronic condition data warehouse, chronic condition categories. Available at: <https://www2.ccmdata.org/web/guest/condition-categories>. Accessed July 1, 2024.
- Jeffery MM, Hooten WM, Henk HJ, et al. Trends in opioid use in commercially insured and Medicare Advantage populations in 2007–16: retrospective cohort study. *BMJ*. 2018;362:k2833.
- Von Korff M, Saunders K, Thomas Ray G, et al. De facto long-term opioid therapy for noncancer pain. *Clin J Pain*. 2008;24(6):521–527.
- Dowell D, Haegerich TM, Chou R. CDC guideline for prescribing opioids for chronic pain—United States, 2016. *JAMA*. 2016;315(15):1624–1645.
- Opioid Oral morphine milligram equivalent (MME) Conversion Factors table for prescription drug coverage; 2020. Available at: <https://www.hhs.gov/guidance/document/opioid-oral-morphine-milligram-equivalent-mme-conversion-factors-0>. Accessed July 1, 2024.
- Mayhew M, DeBar LL, Deyo RA, et al. Development and assessment of a crosswalk between ICD-9-CM and ICD-10-CM to identify patients with common pain conditions. *J Pain*. 2019;20(12):1429–1445.
- Santosa KB, Hu HM, Brummett CM, et al. New persistent opioid use among older patients following surgery: a Medicare claims analysis. *Surgery*. 2020;167(4):732–742.
- Hunnicut JN, Chrysanthopoulou SA, Ulbricht CM, Hume AL, Tjia J, Lapan KL. Prevalence of long-term opioid use in long-stay nursing home residents. *J Am Geriatr Soc*. 2018;66(1):48–55.
- Losina E, Song S, Bensen GP, Katz JN. Opioid use among Medicare beneficiaries with knee osteoarthritis: prevalence and correlates of chronic use. *Arthritis Care Res*. 2023;75(4):876–884.
- Edwards NM, Varnum C, Overgaard S, Nikolajsen L, Christiansen CF, Pedersen AB. Risk factors for new chronic opioid use after hip fracture surgery: a Danish nationwide cohort study from 2005 to 2016 using the Danish multidisciplinary hip fracture registry. *BMJ Open*. 2021;11(3):e039238.
- Oh G, Abner EL, Fardo DW, Freeman PR, Moga DC. Patterns and predictors of chronic opioid use in older adults: a retrospective cohort study. *PLoS One*. 2019;14(1):e0210341.
- Mojtabai R. National trends in long-term use of prescription opioids. *Pharmacoepidemiol Drug Saf*. 2018;27(5):526–534.

35. Gomes T, Campbell T, Tadrous M, Mamdani MM, Paterson JM, Juurlink DN. Initial opioid prescription patterns and the risk of ongoing use and adverse outcomes. *Pharmacoepidemiol. Drug Saf.* 2021;30(3):379–389.
 36. Thornton JD, Dwibedi N, Scott V, et al. Predictors of transitioning to incident chronic opioid therapy among working-age adults in the United States. *Am Health Drug Benefits.* 2018;11(1):12–21.
 37. Udayachalerm S, Bair MJ, Illingworth Plake KS, Huang C-Y, Murray MD, Foster DR. Opioid prescribing and health outcomes in opioid-naïve patients: analysis of a statewide health information exchange. *J Am Pharm Assoc (2003).* 2021;61(5):623–631.e3.
 38. Young JC, Dasgupta N, Chidgey BA, et al. Impacts of initial prescription length and prescribing limits on risk of prolonged postsurgical opioid use. *Med. Care.* 2022;60(1):75–82.
 39. Gomes T, Juurlink DN, Antoniou T, Mamdani MM, Paterson JM, van den Brink W. Gabapentin, opioids, and the risk of opioid-related death: a population-based nested case–control study. *PLOS Med.* 2017;14(10):e1002396.
 40. Bykov K, Bateman BT, Franklin JM, Vine SM, Paterno E. Association of gabapentinoids with the risk of opioid-related adverse events in surgical patients in the United States. *JAMA Netw Open.* 2020;3(12):e2031647.
 41. Simoni AH, Nikolajsen L, Olesen AE, Christiansen CF, Johnsen SP, Pedersen AB. The association between initial opioid type and long-term opioid use after hip fracture surgery in elderly opioid-naïve patients. *Scand J Pain.* 2020;20(4):755–764.
 42. Bigal LM, Bibeau K, Dunbar S. Tramadol prescription over a 4-year period in the USA. *Curr Pain Headache Rep.* 2019;23(10):76.
 43. United States Drug Enforcement Administration. Drug scheduling; 2022. Available at: <https://www.dea.gov/drug-information/drug-scheduling> 2022. Accessed July 1, 2024.
 44. Usmani SA, Hollmann J, Goodin A, et al. Effects of hydrocodone rescheduling on opioid use outcomes: a systematic review. *J Am Pharm Assoc (2003).* 2021;61(2):e20–e44.
 45. Moffat AK, Pratt NL, Kerr M, Kalisch Ellett LM, Roughead EE. Risk of chronic opioid use in older persons with pre-existing anxiety. *J Opioid Manag.* 2019;16(1):59–66.
 46. Daoust R, Paquet J, Moore L, et al. Incidence and risk factors of long-term opioid use in elderly trauma patients. *Ann. Surg.* 2018;268(6):985–991.
 47. Quinn PD, Hur K, Chang Z, et al. Incident and long-term opioid therapy among patients with psychiatric conditions and medications: a national study of commercial health care claims. *Pain.* 2017;158(1):140–148.
 48. Thielke SM, Shortreed SM, Saunders K, Turner JA, LeResche L, Von Korff M. A prospective study of predictors of long-term opioid use among patients with chronic noncancer pain. *Clin J Pain.* 2017;33(3):198–204.
 49. Mosher HJ, Hofmeyer BA, Hadlandsmayth K, Richardson KK, Lund BC. Predictors of long-term opioid use after opioid initiation at discharge from medical and surgical hospitalizations. *J Hosp Med.* 2018;13(4):243–248.
 50. Nadpara PA, Joyce AR, Murrelle EL, et al. Risk factors for serious prescription opioid-induced respiratory depression or overdose: comparison of commercially insured and veterans health affairs populations. *Pain Med.* 2018;19(1):79–96.
 51. Gupta K, Prasad A, Nagappa M, Wong J, Abrahamyan L, Chung FF. Risk factors for opioid-induced respiratory depression and failure to rescue: a review. *Curr. Opin. Anaesthesiol.* 2018;31(1):110–119.
 52. Mues KE, Liede A, Liu J, et al. Use of the Medicare database in epidemiologic and health services research: a valuable source of real-world evidence on the older and disabled populations in the US. *Clin Epidemiol.* 2017;9:267–277.
 53. Hollowell BD, Chambers LC, Barre L, et al. Association between initial opioid prescription diagnosis type and subsequent chronic prescription opioid use in Rhode Island: a population-based cohort study. *BMJ Open.* 2022;12(1):e050540.
 54. Andrews P, Steultjens M, Riskowski J. Chronic widespread pain prevalence in the general population: a systematic review. *Eur J Pain.* 2018;22(1):5–18.
 55. Han B, Compton WM, Blanco C, Crane E, Lee J, Jones CM. Prescription opioid use, misuse, and use disorders in U.S. adults: 2015 National Survey on Drug Use and Health. *Ann Intern Med.* 2017;167(5):293–301.
- Iftexhar Ahmed, BPharm, MS**, PhD Candidate, Department of Pharmacy Practice, College of Pharmacy, Purdue University, Indianapolis and West Lafayette, IN
- Alan J. Zillich, PharmD, FCCP**, Professor, Department of Pharmacy Practice, College of Pharmacy, Purdue University, Indianapolis and West Lafayette, IN; Adjunct Professor, Department of Medicine, School of Medicine, Indiana University, Indianapolis, IN
- Noll L. Campbell, PharmD, MS**, Associate Professor, Department of Pharmacy Practice, College of Pharmacy, Purdue University, Indianapolis and West Lafayette, IN; Adjunct Associate Professor, Department of Medicine, School of Medicine, Indiana University, Indianapolis, IN; Investigator, Indiana University Center for Aging Research, Regenstrief Institute Inc, Indianapolis, IN
- Kevin M. Sowinski, PharmD, FCCP**, Professor, Department of Pharmacy Practice, College of Pharmacy, Purdue University, Indianapolis and West Lafayette, IN; Adjunct Professor, Department of Medicine, School of Medicine, Indiana University, Indianapolis, IN
- David R. Foster, PharmD, FCCP**, Associate Professor, Department of Pharmacy Practice, College of Pharmacy, Purdue University, Indianapolis and West Lafayette, IN; Adjunct Associate Professor, Department of Medicine, School of Medicine, Indiana University, Indianapolis, IN